Problem Statement – Leo Rover Mission

The central objective of this engineering project is to design, implement, and validate a fully autonomous mobile manipulation platform built upon the Leo Rover mobile base. The system is required to operate independently within a bounded, dynamic arena to locate, pick up, transport, and accurately sort discrete color-coded blocks into designated target bins located near the starting baseline. All operational tasks—spanning system initialization, environmental exploration, object manipulation, block deposition, and final task termination—must be executed seamlessly and completely autonomously within a strict 20-minute operational window.

Autonomous mobile manipulation represents a critical frontier in modern robotics, bridging the operational gap between stationary industrial manipulators and unassisted mobile rovers. The principal engineering challenge stems from operating in an unstructured, non-deterministic environment where arena boundaries, block spatial coordinates, and target bin positions are randomized prior to testing. As a result, the platform cannot utilize mapped environments, fixed way points, offline global route planning, or prerecorded manipulator arm trajectories.

To achieve mission success, the mobile robot must reliably execute real-time simultaneous localization and mapping (SLAM), dynamically identify and track target blocks by color under variable ambient lighting conditions, generate collision-free navigation paths around static and moving obstacles, and execute precise arm kinematics for repeatable block retrieval and sorting. Ultimately, the system must seamlessly integrate structural hardware modifications, onboard spatial computing, real-time computer vision processing, sensor fusion, and closed-loop kinematic control into a robust, high-performance autonomous platform capable of reliable real-world deployment.